

YVONNE MONIQUE LOPEZ DAVILA

Boston, MA | ymld2021@gmail.com | [linkedin.com/in/ymld975](https://www.linkedin.com/in/ymld975) | yvonne-lopez-davila.github.io

EDUCATION

Tufts University Medford, MA

M.S in Software Systems Development

GPA: 4.00 | Expected Dec 2026

B.S in Computer Science & Quantitative Economics

GPA: 3.79 | Magna Cum Laude | Spring 2025

Courses: Distributed Systems, Cloud Computing, Big Data, AI for Social Impact, Algorithm/Data Structs, Security, ML

SKILLS

Programming Langs: C++, C, Python (NumPy, Pandas), JavaScript, Java, Stata, C#, HTML, SQL, MATLAB

Tools & Methods: Docker, Git, Linux, Hadoop, Spark, MapReduce, LangChain, Agile Dev, MS Excel, GCP

Languages: English (*fluent*), Spanish (*fluent*), French (*prof proficiency*), Italian (*lim prof proficiency*)

EXPERIENCE

The MITRE Corporation Bedford, MA, *Software Engineering Intern*

Incident Management and Emergency Response Application

Summer 2024, 2025

- Improved flexibility & scalability of AI chatbot through **architectural redesign** that introduced an **API microservice layer**, enabled **persistent session data**, & **integrated AI bot into MS Teams**
- Designed validation algorithm** to score AI responses & identify hallucinations
- Transformed & sorted input** using 2D-trees to facilitate $O(\log n)$ alert transmissions & **visualization** of cumulative time-series/qualitative data in local web app

Planning and Development: Full stack web-application

Summer 2023

- Collaborated in Scrum team to iteratively develop a **full-stack application** that enabled users to access & update mission data via geospatial queries on an **interactive map** (*REACT, Typescript*)
- Developed back-end logic (C#, .NET) to **read, process, & store geo-indexed data**
- Created **controller-based web API** to manage metadata stored in MongoDB

Tufts University Medford, MA, *Discrete Math Course Assistant*

2 semesters – 2023, 2025

PROJECTS

Combating LA Wildfires via GenAI: Home Risk Assessment & Recommendation System

April 2026

Tufts x Microsoft AI for Good

- Built a **full-stack, LLM-powered risk assessment** system that uses **RAG retrieval** and **trend analysis** to generate structured insights from geospatial, structural, and insurance datasets
- Designed **role-specific workflows** for buyers and owners that enabled **tailored, actionable steps** through ranked risk insights or mitigation checklists, respectively
- Conducted a **randomized controlled experiment** to **measure system performance** and usefulness relative to existing, general AI systems

End-to-End File Copying Program Over a Simulated Network | C++ | *Systems* | *UDP*

Oct 2025

- Developed a reliable system to copy client files to server directories **over unreliable UDP sockets**
- Implemented **retransmission logic** and **end-to-end verification** to ensure **fault tolerant delivery** under simulated packet loss, delays, and reordering

Cancerous Skin Lesion Image Classifier | *ML* | *Python* | *Computer Vision*

May 2026

- Built an ensemble-based ML pipeline combining image and tabular patient data to classify lesions requiring treatment (0.946 AUROC on held-out test data)
- Designed a custom fairness metric to reduce age-based performance biases
- Implemented ensemble model combining transfer learning techniques with *CatBoost* gradient boosting (Bayesian hyperparameter search)

Remote Procedure Call Framework | C++ | *IDL Parsing* | *Code Generation*

Nov 2025

- Implemented an RPC system that **autogenerates client stubs and server proxies** from function interface definition files to enable **remote execution** of function calls